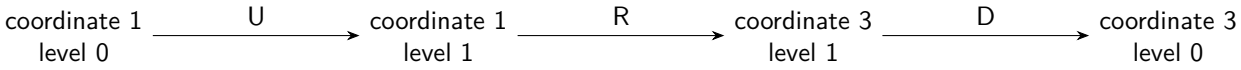
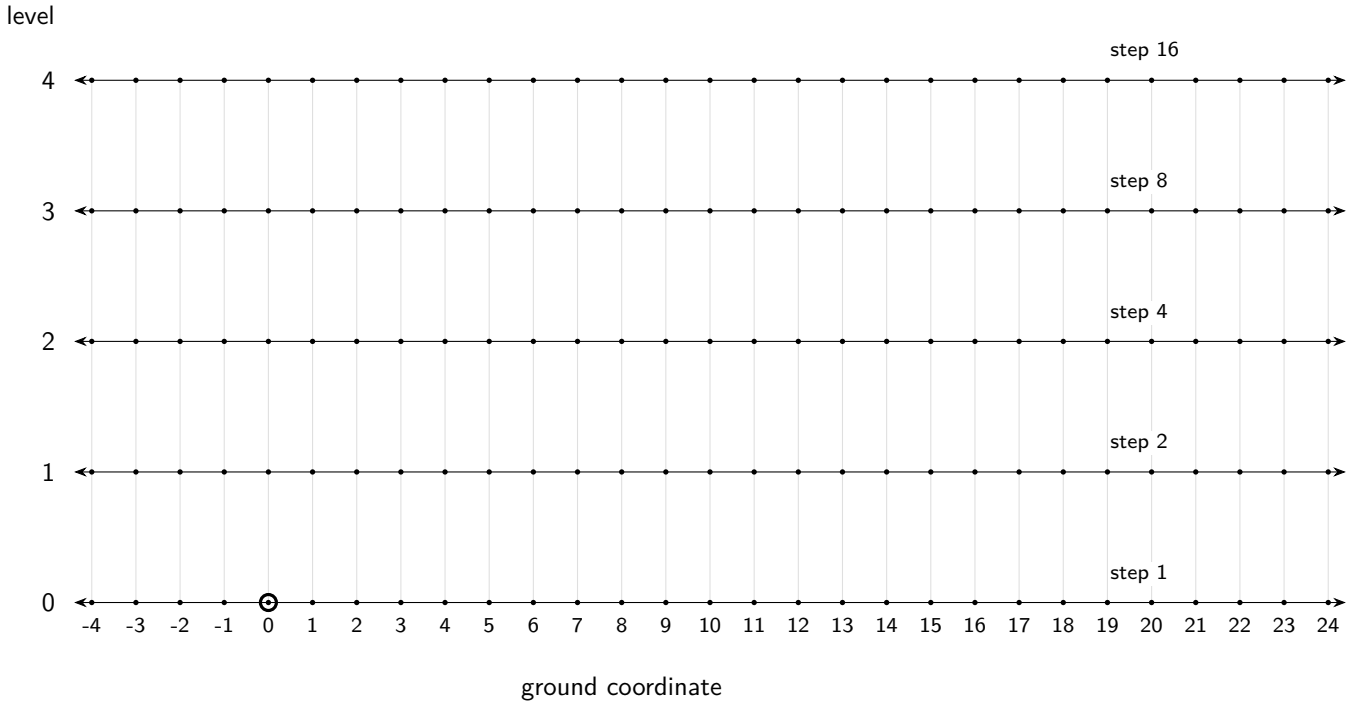
Move a marker. Your partner checks each move and writes its letter. Swap jobs after a trip. Start at ground coordinate 0, level 0.

U and D move one level straight up or down, keeping the ground coordinate. Do not go below level 0. R and L move right or left by the level's step size. Every move costs one. The roads continue beyond this page; higher levels keep doubling.



Example: U R D costs 3 moves.

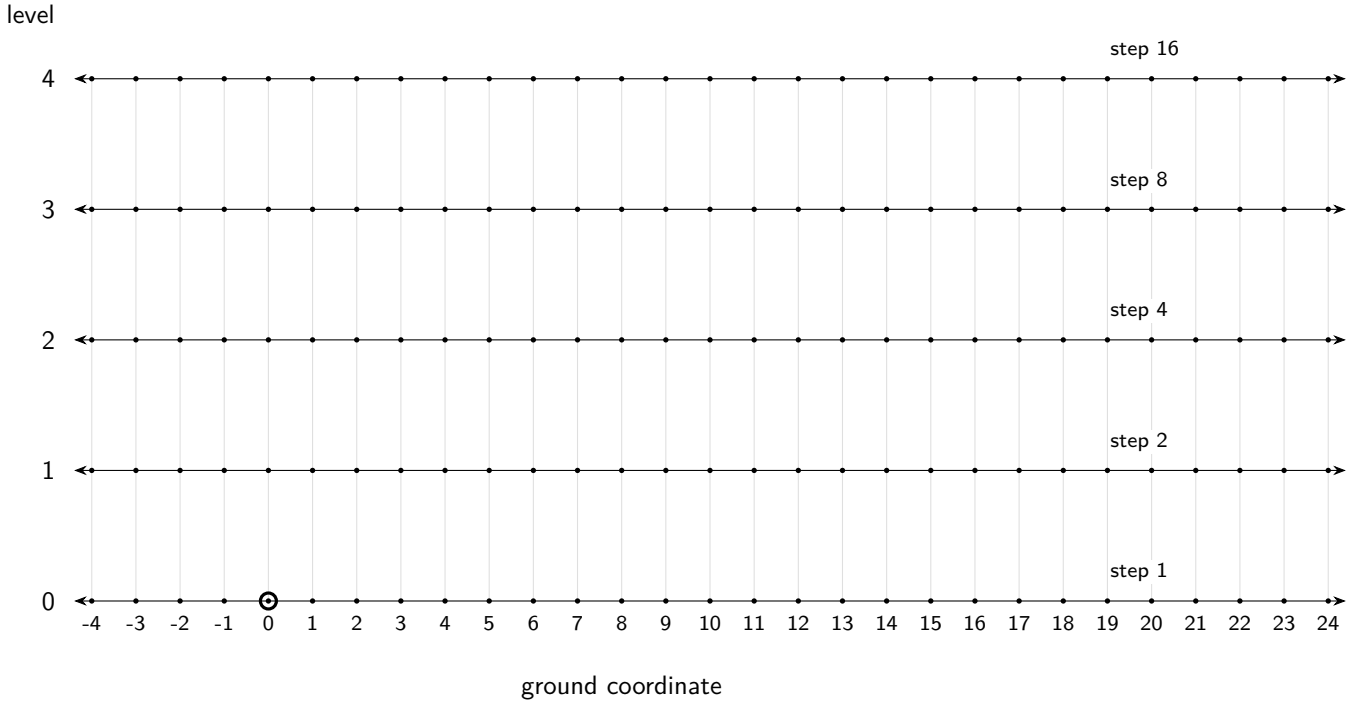
Problem 1: Reach coordinate 3 and return to level 0 in as few moves as you can. Then try coordinate 7. Your partner may challenge a route with a shorter one.



3: _____

7: _____

Problem 2: Reach coordinate 16 and return to level 0. Find the shortest trip you can. Can you find two different trips tied for your best?



Move letters

Number of moves

_____

_____

_____

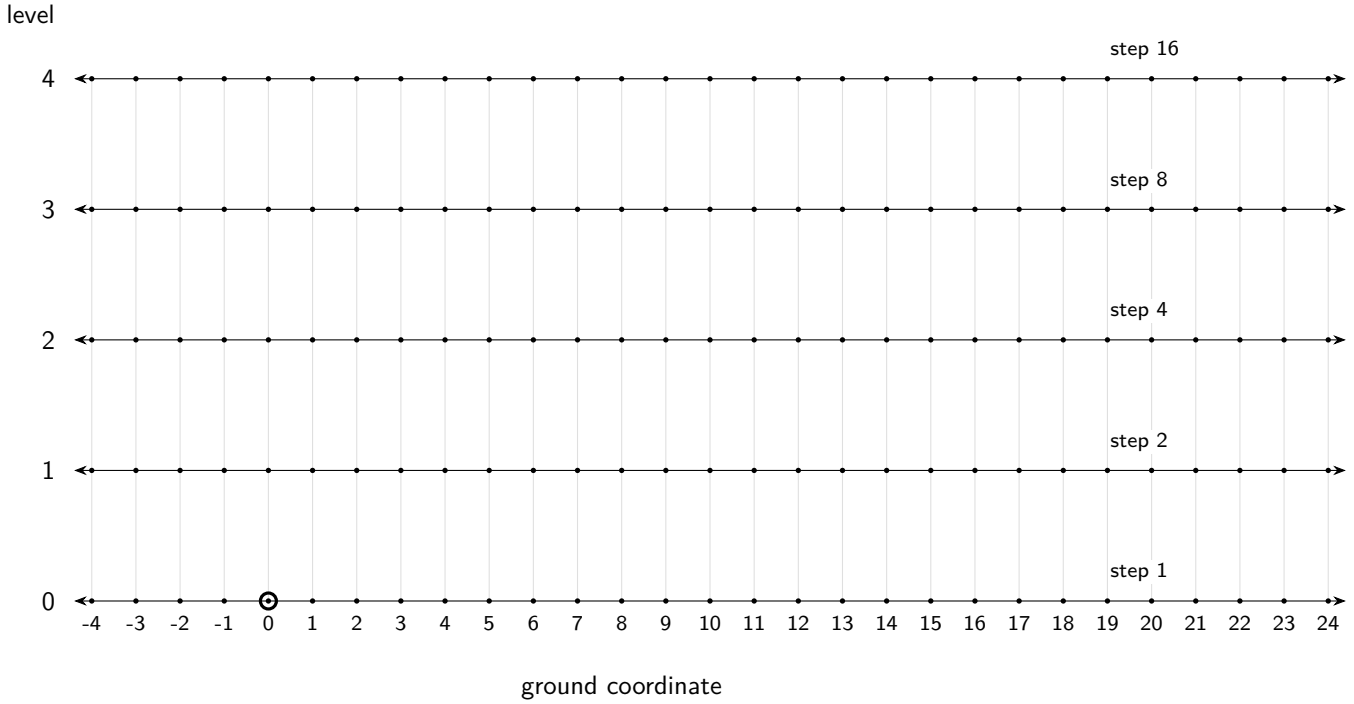
Shortest trip so far: _____ moves.

Problem 3: With each move budget, finish as far right as possible on level 0. Your partner tries to beat your distance. You may use fewer moves than the budget.

Budget	Move letters	Farthest coordinate
4		
5		
6		
7		
8		

Problem 4: Can any trip reach 16 and return to level 0 in seven or fewer moves? Explain why your answer covers trips that move left or change levels many times.

Problem 5: Find the shortest trips you can to 9, 15, 17 and 23, ending on level 0. Can a trip that moves left beat every trip that only moves right, up and down? Choose a destination and settle this with your partner.



9: _____

15: _____

17: _____

23: _____